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Calibration of Laser Doppler Anemometer: Fringe Spacing and Velocity

F S Ferreira¹, F O Costa¹, I P Barros¹, S Araújo¹

¹ National Institute of Metrology, Quality and Technology – Inmetro
 Directory of Scientific and Technology Metrology - Dimci
 Fluid Dynamics Metrology Division – Dinam
 Gas Flow Laboratory - Lagas
 Av. Nossa Senhora das Graças, 50 – Prédio 6 - Duque de Caxias – RJ
 E-mail: fsferreira@inmetro.gov.br

Abstract. This paper presents the most recent mathematical modelling and procedure for Laser-Doppler Anemometers (LDA) calibration used by the Brazilian National Institute of Metrology Institute, the National Institute of Metrology, Quality and Technology (Inmetro). Here we calculate the fringe spacing calibration coefficient (C_f) and the Burst Spectrum Analyser (BSA) calibration coefficient (C_{bsa}). Using these two calibration coefficients we can calibrate a LDA in a determinate velocity range.

Keyword. Calibration; LDA; Spacing; Fringe; Velocity; Anemometry.

1. Introduction

Most of the National Metrology Institutes use a calibration disk as a primary standard for fluid speed [1]. The working principle is based on the calculation of the linear speed of a rotating disk and the direct comparison with the speed in a Laser Doppler anemometry system. The LDA readings detect some particle moving on the disk, which may be valleys and peaks of the surface roughness [2], a tightly bound tungsten wire [3, 4], or particles embedded in the building material of a glass disk [5, 6].

The LDA calculates the speed by determination of the Doppler frequency, reflected by a particle passing through the control volume under non-slip condition [7]. Knowing the spacing between the fringes, a linear relationship can be easily deduced to calculate fluid speed. The spacing between the fringes can be theoretically determined according to the equation (a.1):

$$\delta = \frac{\lambda}{2\sin(\theta/2)} \quad (\text{a.1})$$

Where λ is the wavelength of the laser beam and θ is the angle of intersection.

However, imperfections of lens, slight variations in wavelength and phase shift outside the laser beam waist create distortions in the fringe spacing, making it necessary to characterize the control volume.



As the distance between interfering fringes are independent of the fluid speed, by calibrating the Burst Spectrum Analyzer in frequency, the entire range of velocities can be determined by the equation (a.2):

$$v = \delta \cdot f_d \quad (\text{a.2})$$

2. Development

2.1. Calibration of the Speed standardization Rotation Disk

The rotation disk was calibrated by Inmetro's Dimensional Laboratory. The diameter was calibrated in three different heights (figure 1). To each height eight diameters in different directions and the out of roundness were measured.

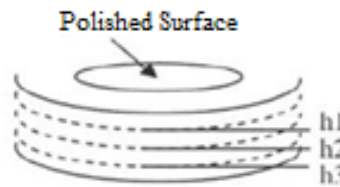


Figure 1 – Measurements Heights

For the purpose uncertainty calculation, the reference diameter can be determined as the mean of all calibrated diameters and the out of roundness can be determined as the random variation of the diameter:

$$d_0 = \frac{1}{n \cdot m} \sum_{i=1}^n \sum_{j=1}^m d_{i,j} + \delta d \quad (\text{a.3})$$

Where n is the number of measured diameters at each one of the m heights and δd is the random variation of the diameter, which can be calculated as the mean out of roundness:

$$\delta d = \frac{1}{m} \sum_{j=1}^m \delta d_{i,j} \quad (\text{a.4})$$

2.2. Reference Speed

The reference speed for the LDA calibration shall be the tangential velocity of the disk at the intersection position between the control volume and the disk surface. Figure 2 shows the used coordinate system.

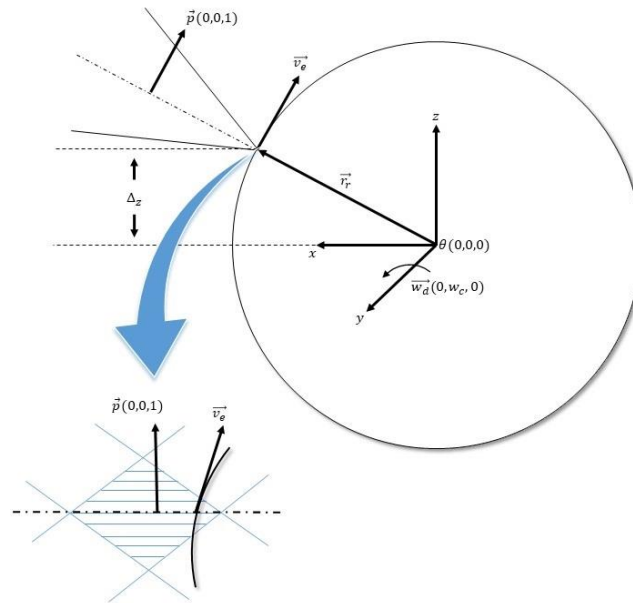


Figure 2 – Coordinate System

The reference for calibration will be the plane perpendicular to the plane formed by the laser beams, which can be described by a vector:

$$\vec{p} = \langle 0, 0, 1 \rangle \quad (1)$$

This vector will be used as the basis for calculating the alignment between the control volume and the calibration disk. The angular velocity of the calibration disk can also be described by a vector:

$$\vec{\omega}_d = \langle 0, \omega_c, 0 \rangle \quad (2)$$

Where ω_c is the calibrated angular speed of the calibration disk, and:

$$V_e = \langle \sqrt{\left(\frac{d_d}{2}\right)^2 - \Delta z^2}, 0, \Delta z \rangle \quad (3)$$

The point where the Control Volume intercepts the surface of the calibration disk, where d_d is the disk diameter and Δz is the vertical alignment error. The arc length at V_e is:

$$\vec{r}_r = \langle 0, 0, 0 \rangle - V_e \quad (4)$$

The surface velocity will be given by the cross product between $\vec{\omega}_d$ and $\vec{r}_r$:

$$\vec{v}_l = \vec{\omega}_d \times \vec{r}_r = \langle -\Delta z \omega_d, 0, -\sqrt{\frac{d_d^2}{4} - \Delta z^2} \omega_d \rangle \quad (5)$$

The speed seen by the LDA will be given by the projection of $\vec{v}_l$ in $M \cdot \vec{p}$, where M is the rotation matrix which describes angular misalignment of the system:

$$M = M_z \cdot M_x \cdot M_y \quad (6)$$

$$M_x = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos(\phi x) & \sin(\phi x) \\ 0 & -\sin(\phi x) & \cos(\phi x) \end{bmatrix} \quad (7)$$

$$M_y = \begin{bmatrix} \cos(\phi y) & 0 & -\sin(\phi y) \\ 0 & 1 & 0 \\ \sin(\phi y) & 0 & \cos(\phi y) \end{bmatrix} \quad (8)$$

$$M_z = \begin{bmatrix} \cos(\phi z) & -\sin(\phi z) & 0 \\ -\sin(\phi z) & \cos(\phi z) & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad (9)$$

By the equation (5) we have:

$$V_{ref} = \frac{\omega_d}{2} \left| \sqrt{d_d^2 - 4\Delta z^2 \cos(\phi x) \cos(\phi y) + 2\Delta z \sin(\phi x) \sin(\phi z) \cos(\phi y)} \right. \\ \left. + 2\Delta z \cos(\phi z) \sin(\phi y) \right| \quad (10)$$

$$\omega_d = 2\pi f_d \quad (11)$$

$$f_d = \frac{f_c}{n_p + \delta n_p} \quad (12)$$

$$d_d = (d_0 + \delta d_0)(1 + \alpha(T_m + \delta T_m - T_r)) \quad (13)$$

$$T_m = \frac{T_1 + T_2}{2}, \quad (14)$$

Where:

ω_d	angular velocity of the calibration disk;
Δz	vertical alignment error;
$\phi x, \phi y, \phi z$	system angular alignment errors;
d_0	calibrated diameter of the disk (mean of the diameters);
f_d	frequency of the calibration disk;
f_c	clock frequency of the time and frequency pattern;
n_p	number of mean clock pulses per revolution;
δn_p	error due to random variation of the number of clock pulses;
δd_0	error due to the mean circularity of the disk;
α	coefficient of linear expansion of steel;
T_m	temperature during measurements;
T_{1i}, T_{2i}	initial temperature of the disk at spots 1 and 2, respectively;

T_{1f}, T_{2f}	final disk temperature at points 1 and 2, respectively;
δt_m	error due to random temperature variation during measurements; and
T_r	reference temperature (obtained on the calibration certificate of the disk diameter).

2.3. Calculation of the Calibration Constant for the fringe Spacing (C_f)

The calibration coefficient for fringe spacing can be determined as the ratio between the reference velocities and the reconstructed speed of the LDA. The speed of the LDA needs to be reconstructed because, during calibration, only one position of the control volume can be observed. In real flows, particles pass homogeneously throughout its entire length. However, the probability of a particle being detected is not uniform throughout the control volume. Thus, the control volume should not only be characterized as to its distortion, but also as to the relative acquisition rate at a given point [11]. The reconstructed speed is calculated as the average speed weighted by the Data Rate (or by the number detected values) in each position:

$$C_f = \frac{v_{ref}}{v'_{lda}} + \delta C_f \quad (15)$$

$$v'_{lda} = \frac{\sum_{i=1}^N (v_i + \delta v_i) n_i}{\sum_{i=1}^N n_i} \quad (16)$$

Where:

C_f	calibration constant;
v_{ref}	reference speed (linear velocity of the calibration disk);
v'_{lda}	speed calculated by scanning the control volume;
δC	error due to repeatability;
N	number of positions measured;
v_i	mean speed of i -th measured position;
δv_i	random variation of the speed of i -th measured position; and
n_i	number of points measured in the i -th measured position.

2.4. Calculation of the BSA constant (C_{BSA})

The Burst Spectrum analyzer shall be calibrated for frequency measurement to achieve traceability at speed over the entire desired range. However, many LDA systems do not have an input channel directly connected to the data acquisition board, and this calibration cannot be performed directly by a time and frequency laboratory.

In this way, the calibration disk can be used again to perform this step. Due to the symmetry of the control volume for the acquisition rate and the approximately linear variation of the distortion with the position, the calibration coefficient C_f will be adjusted for the Doppler frequency obtained at the center (position of the highest data rate) of the control volume. This position can be determined through the two highest data rate points. Let these positions be x_1 and x_2 with data rates of Dr_1 and Dr_2 , respectively. The highest data rate (x_{max}) position, where the BSA calibration will be performed, will be:

$$x_{max} = \frac{Dr_1 x_1 + Dr_2 x_2}{Dr_1 + Dr_2} \quad (17)$$

By determining a point in the control volume with a high acquisition rate and by varying the frequency of the disk, signals with different Doppler frequencies will be generated, allowing the calibration of the frequency card. The direct relationship between the reference speed and the speed obtained in v_{lda} determines a BSA coefficient for a given velocity:

$$C'_{bsa} = \frac{v_{ref}}{v_{lda} + \delta v_{lda}} + \delta C'_{bsa} \quad (18)$$

It is observed in the equation above that the LDA velocity doesn't need to be reconstructed as in the fringe spacing calibration, since this step is only intended to determine the BSA frequency card. The superscript ' ' in the calibration coefficient of the BSA is relative to the unadjusted constant. This adjustment is necessary so that the fringes distortion is not considered twice:

$$C_{bsa} = \frac{C'_{bsa}}{C'_{bsa,f}} \quad (19)$$

Where:

C'_{bsa}	unmatched constant;
v_{lda}	speed read by LDA;
$\delta C'_{bsa}$	error due to random variation of the unbalanced constant;
$C'_{bsa,f}$	constant not compensated for the speed used during the calibration of the spacing between stripes; and
δv_{lda}	error due to the random variation of the speed read by the LDA.

Strictly speaking, an iterative process between the calibration of the spacing between the fringes and the calibration of the BSA should be performed: In performing the calibration of the spacing between the fringes, the differences between the speeds at each position of the control volume should be corrected by the calibration of the BSA. However, the differences between the frequencies are extremely small relative to the stability of the BSA frequency card, so that its coefficient can be considered constant for values around the calibrated frequency.

2.5. Calibrated speed calculation

The speed of the LDA is determined by multiplying the two calibration coefficients at the speed read by the LDA system:

$$v_c = C_{bsa} C_f v_{lda} \quad (20)$$

2.6. Automation and data acquisition

System automation and data acquisition was developed in *LabView* environment. Due to the high frequency of the reference clock, the angular frequency measurement of the disk was developed in FPGA logic and compiled in NI cRIO-9073 controller, with data transfer using FIFO structure, as shown in figure 3:

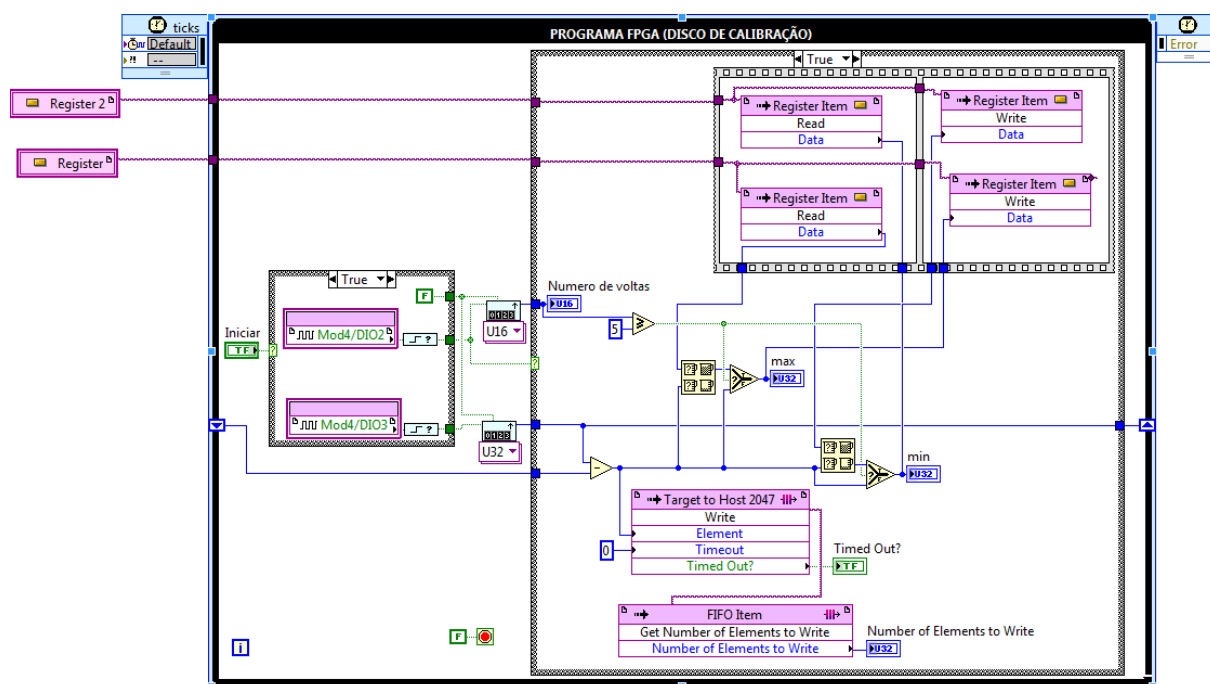


Figure 3 - Program Logic

The programming in FPGA is essential for precisely measure the period of rotation of the calibration disk. A 32-bit counter integrates the number of pulses of the time reference, with the beginning and end of the counts determined by the signal coming from the photodetector circuit. In this way, each revolution generates a value of number of pulses and, consequently, period of rotation. The signals are stored in memory, forming a data queue with FIFO structure that is downloaded to the user interface.

A user interface was also created in *LabView*, operating in *Windows* environment. After the end of a measurement, the data is exported in a text file.

2.7. Results and Uncertainty calculation

The uncertainty calculation was performed according to “Evaluation of Measurement Data — Guide to the Expression of Uncertainty in Measurement” (ISO-GUM) [9] in *Excel* worksheet and validated in *Mathematica* software. [8].

The final uncertainty of the calibration was 0.2%, with major contributions of the repeatability (56%), alignment (20%) and the diameter calibration (16%).

3. Conclusions

The uncertainty of the system showed good agreement to other National Institutes of Metrology. The calibration of the BSA can lead to virtually unlimited speed range, requiring only fringe spacing calibration and a frequency based calibration of the frequency card.

Developments of mathematical models for compensating temperature effects on the probe and lenses can enhance the repeatability of the calibration, while the assessing the effects of the laser intensity and its intensity distribution can be useful for improving the data rate and velocity profile.

The calibration of Laser Doppler Anemometers is the first step to create the traceability chain of fluid speed. Next steps include calibration of Ultrasonic anemometers and Pitot Tubes.

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